

2.7 Propositional Satisfiability

Propositional satisfiability checks whether a statement can be true. It tells us if a logical expression has at least one situation where it is true.

Where we use Propositional Satisfiability?

- Widely used in computer science, especially in problem solving and programming.
- Helps test logical statements and make decisions.
- Important for designing algorithms and verifying programs.
- It helps in solving complex problems efficiently.
- It helps in improving reasoning skills.

Example:

- “Bring pizza OR (burger AND do NOT bring soda)” → Satisfiable
- “You must bring pizza AND you must NOT bring pizza” → Unsatisfiable

DO YOU KNOW? A logical expression is satisfiable if there is at least one way to make it true.

Conditions for a Proposition to be Satisfiable

- A proposition is **satisfiable** if there is at least one combination of truth values that makes it true. If all possible combinations make it false, it is **unsatisfiable**.
- Satisfiability depends on the structure of the expression and the logical connectives used.
- Checking all possible cases using a truth table is one way to test satisfiability.

Examples of Satisfiable and Unsatisfiable Expressions

- Satisfiable: $P \text{ OR } Q$ (true if either P or Q is true)
- Satisfiable: $P \text{ AND } Q$ (true when both are true)
- Unsatisfiable: $P \text{ AND NOT } P$ (never true)

Short Questions

1. What is propositional satisfiability?
2. When is a proposition called satisfiable?
3. When is a proposition called unsatisfiable?
4. Give one example of a satisfiable expression.
5. Give one example of an unsatisfiable expression.
6. How can we check whether a proposition is satisfiable?
7. Why is satisfiability important in computer science?
8. What is the difference between satisfiable and unsatisfiable propositions?